

3.4 Independence and zero covariance

For any random vectors with finite second moments, independence implies zero cross-covariance, because the expectation of a product of functions of independent variables factors. The converse fails in general, as Example (A random sign) and Example (A mixture of two correlations) show. For jointly normal vectors it holds. This is the single most used property of the normal distribution in linear-model theory. Every independence statement behind a t or F statistic reduces to it.

Theorem 3.4.1 — Independence

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be partitioned into blocks $\mathbf{Y} = (\mathbf{Y}_1^\top, \dots, \mathbf{Y}_k^\top)^\top$, with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_{ij})$ partitioned to match.

- (a) The blocks $\mathbf{Y}_1, \dots, \mathbf{Y}_k$ are mutually independent iff $\boldsymbol{\Sigma}_{ij} = \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j) = \mathbf{0}$ for all $i \neq j$.
- (b) In particular, $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $\text{Cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{0}$, and the scalar entries $Y_1, \dots, Y_n$ are mutually independent iff $\boldsymbol{\Sigma}$ is diagonal.

Proof. If the blocks are independent, every cross-covariance is zero. Conversely, suppose $\boldsymbol{\Sigma}_{ij} = \mathbf{0}$ for $i \neq j$, and partition $\mathbf{t}$ to match. Then

$$\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} = \sum_{i=1}^k \left(\mathbf{t}_i^\top \boldsymbol{\mu}_i + \frac{1}{2} \mathbf{t}_i^\top \boldsymbol{\Sigma}_{ii} \mathbf{t}_i \right),$$

because all the cross terms $\mathbf{t}_i^\top \boldsymbol{\Sigma}_{ij} \mathbf{t}_j$ vanish. Exponentiating, Theorem 3.2.1 gives $M_{\mathbf{Y}}(\mathbf{t}) = \prod_i M_i(\mathbf{t}_i)$, where M_i is the moment generating function of $N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$. By Corollary 3.3.2 that is the distribution of $\mathbf{Y}_i$, so $M_i = M_{\mathbf{Y}_i}$. The factorization property of Section 3.2 gives mutual independence. Part (b) is the cases $k = 2$ and $k = n$. ■

A consequence worth stating explicitly: for jointly normal blocks, *pairwise* independence implies *mutual* independence, because condition (a) involves only pairs. For general random vectors this is false, as the example of three random signs in Example (Pairwise but not mutually independent) shows. Joint normality of the whole vector matters here: joint normality of each pair is not enough (Exercise (C1)).

The theorem is usually applied to two linear functions of the same normal vector.

Corollary 3.4.2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $\mathbf{A}$ and $\mathbf{B}$ be matrices with n columns. Then $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$. In that case $g(\mathbf{AY})$ and $h(\mathbf{BY})$ are independent for all measurable functions g and h .

Proof. The stacked vector $((\mathbf{AY})^\top, (\mathbf{BY})^\top)^\top$ is a linear function of $\mathbf{Y}$, so it is normal by Theorem 3.3.1. Its off-diagonal covariance block is $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$

(Theorem 2.2.1). Apply [Theorem 3.4.1](#). Functions of independent random vectors are independent. ■

The condition $\mathbf{A}\Sigma\mathbf{B}^\top = \mathbf{0}$ involves only second moments, but the conclusion is full independence. In particular, any quadratic form in $\mathbf{BY}$, such as $\|\mathbf{BY}\|^2$, is independent of $\mathbf{AY}$. [Chapter 4](#) extends this to quadratic forms that are not written as squared lengths (Theorem 4.4.1).

Example. (Sample mean and deviations)

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$, so that $Y_1, \dots, Y_n$ is a random sample from $N(\mu, \sigma^2)$. The mean is $\bar{Y} = n^{-1}\mathbf{1}^\top\mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$. As in Example (Centring a sample),

$$\text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top,$$

By [Corollary 3.4.2](#), $\bar{Y}$ is independent of $\mathbf{d}$, and so of every function of the deviations: the sample variance $S^2 = \|\mathbf{d}\|^2/(n-1)$, the range, the sample skewness. The distribution of $(n-1)S^2/\sigma^2$ is found in Corollary 4.4.5. The computation used $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ only through $\Sigma\mathbf{1} \in \text{span}(\mathbf{1})$, which is the precise condition for the conclusion ([Exercise \(B1\)](#)).

Example. (Coefficients and residuals)

In the setting of Example (Least squares under normal errors),

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \mathbf{I}) (\mathbf{I} - \mathbf{M}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \{(\mathbf{I} - \mathbf{M})\mathbf{X}\}^\top = \mathbf{0},$$

since $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$. Hence $\hat{\boldsymbol{\beta}}$ is independent of the residual vector, and so of the residual sum of squares $\|\hat{\boldsymbol{\varepsilon}}\|^2$ and of the estimate $s^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2/(n-p)$. The same argument shows that the fitted values $\hat{\mathbf{Y}}$ and the residuals are independent. This independence is what allows a t statistic $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\{s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}\}^{1/2}$ to have a t distribution (Theorem 11.6.1).

Remark (What joint normality buys). The two examples use only a covariance calculation. Without joint normality, zero covariance between $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ still holds under the second-moment assumptions of Definition 5.3.1, but independence does not follow, and neither do exact t and F distributions. [Chapter 19](#) (Theorem 19.3.2) and [Chapter 21](#) examine what survives approximately.

3.4.1 Exercises

3.4.1.1 A. Check your understanding

Exercise A1.

Let Y_1, Y_2 be independent $N(\mu, \sigma^2)$. Show that $Y_1 + Y_2$ and $Y_1 - Y_2$ are independent. What if the variances differ?

Exercise A2.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{u}$ be a unit vector. Show that $\mathbf{u}^\top \mathbf{Y}$ and $\|\mathbf{Y} - (\mathbf{u}^\top \mathbf{Y})\mathbf{u}\|^2$ are independent.

3.4.1.2 B. Practice**Exercise B1.**

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \Sigma)$ with Σ arbitrary. Show that $\bar{Y}$ is independent of the deviations $Y_i - \bar{Y}$ iff $\Sigma\mathbf{1} \in \text{span}(\mathbf{1})$, that is, iff all row sums of Σ are equal. Check the condition for the equicorrelation matrix $\sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$, and for a two-group matrix: the first m_1 and the last m_2 observations form groups, all variances are 1, the correlation is ρ within a group and τ between groups.

Solution. $\bar{Y} = n^{-1}\mathbf{1}^\top \mathbf{Y}$ and the deviations are $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$, so by [Corollary 3.4.2](#) they are independent iff $n^{-1}\mathbf{1}^\top \Sigma(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top$, that is, $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\Sigma\mathbf{1} = \mathbf{0}$. The null space of $\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ is $\text{span}(\mathbf{1})$, and $\Sigma\mathbf{1}$ is the vector of row sums. For the equicorrelation matrix, $\Sigma\mathbf{1} = \sigma^2(1 - \rho + n\rho)\mathbf{1}$, so independence holds for every admissible ρ . In the two-group matrix, a row in the first group sums to $1 + (m_1 - 1)\rho + m_2\tau$ and a row in the second to $1 + (m_2 - 1)\rho + m_1\tau$. The difference is $(m_1 - m_2)(\rho - \tau)$, so independence holds iff the groups have equal size or $\rho = \tau$, and the latter is the equicorrelated case.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\text{rank}(\mathbf{X}) = r < p$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$. Show that the least squares estimate $\mathbf{G}\mathbf{X}^\top \mathbf{Y}$ and the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{Y}$ are independent, and that every estimable function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ has an estimate independent of the residual sum of squares.

3.4.1.3 C. Going deeper**Exercise C1.**

Let φ be the $N(0, 1)$ density, $0 < \varepsilon \leq e^{3/2}$, and

$$f(x, y, z) = \varphi(x)\varphi(y)\varphi(z)\{1 + \varepsilon xyz e^{-(x^2+y^2+z^2)/2}\}.$$

Show that f is a probability density, and that under f every pair of coordinates is $N_2(\mathbf{0}, \mathbf{I})$. So the three variables are pairwise jointly normal and pairwise independent. Show that they are not mutually independent and that (X, Y, Z) is not jointly normal. Which step of [Theorem 3.4.1](#) fails?

Solution. Since $|t|e^{-t^2/2} \leq e^{-1/2}$ for all t , the perturbation satisfies $|\varepsilon xyz e^{-(x^2+y^2+z^2)/2}| \leq \varepsilon e^{-3/2} \leq 1$, so $f \geq 0$. The function $z \mapsto z\varphi(z)e^{-z^2/2}$ is odd and integrable, so integrating f over z removes the perturbation and leaves $\varphi(x)\varphi(y)$. Integrating further shows that f integrates to 1, and by symmetry each pair has density $\varphi \otimes \varphi$, the $N_2(\mathbf{0}, \mathbf{I})$ density. Mutual independence would force $f = \varphi(x)\varphi(y)\varphi(z)$ almost everywhere, but the perturbation is nonzero off the coordinate planes. Directly, $E(XYZ) = \varepsilon(\int t^2 \varphi(t) e^{-t^2/2} dt)^3 > 0$,

whereas independence with mean-zero margins gives 0. If (X, Y, Z) were jointly normal, its covariance matrix would be $\mathbf{I}$ (unit variances from the margins, zero covariances from the pairs), and by [Theorem 3.4.1](#) the coordinates would be independent, which they are not. The theorem's hypothesis is joint normality of the whole vector. Here only the pairs are normal, and the moment generating function of the triple does not factor.